

D871

Secondary Chemistry Curriculum

Armored Drill

8 concepts · 12 items · 4 transfer items

What this is. Every concept in this course's public description, re-asked with more disguise than a practice test uses. Each stem carries facts that do not decide it. Every wrong answer is a named trap.

Source. Scope from the course's public description in the WGU Institutional Catalog (September 2026); items written from the field's standard body of knowledge. No WGU assessment item, course material or proprietary content is used or reproduced. Not affiliated with or endorsed by Western Governors University.

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How to run it

Paper first. Pencil. One pass, no notes, a timer if you want the pressure.

1. **Before you read the options**, fill the strip under the item: the target word (what is actually being asked), and the one fact in the stem that decides it.
2. **Principle before arithmetic**. Name the rule, then compute.
3. **Camouflage**. Most of every stem does not matter. Ask: which single fact decides this?
4. **Inverse pairs are the bait**. If two options are opposites, one of them is the trap built for you.
5. **Mark sure, unsure or guess** honestly. Sure-and-wrong is the most valuable miss on the page — it is a wrong mental model.
6. **Absolutes** (always, never, guarantee) are usually wrong. And recognizing an answer is not the same as owning it.
7. Items with a **heavy black bar** on the left are **transfer items**: the same principle, reversed or solved for a different variable.
8. Stop at the STOP page. Then grade, fill the triage grid, and tally the traps you fell for. The biggest tally is what you drill next.

The strip, filled in, looks like this:

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	_____	

Items

1.

While planning an atomic-structure lesson for a tenth-grade class that meets after lunch, a teacher candidate uses boron: 20.0% is boron-10 (10.0 u) and 80.0% is boron-11 (11.0 u). The periodic-table poster in the room is faded. What average atomic mass should the answer key show?

- A. 11.0 u
- B. 10.8 u
- C. 10.2 u
- D. 10.5 u

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	_____	

2. [TRANSFER — reversed stem / different variable]

An element has two isotopes, 63 u and 65 u, and an average atomic mass of 63.5 u. What percent of its atoms are the 63 u isotope?

- A. 75%
- B. 50%
- C. 63.5%
- D. 25%

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	_____	

3.

In a lab station set up next to the room's only sink, a white crystalline solid melts at 801 °C. It does not conduct electricity as a solid, but it conducts well when dissolved in water. The solid came from a kit ordered from a discount supplier. What type of bonding does it have?

- A. Ionic
- B. Molecular covalent
- C. Metallic
- D. Covalent network, because anything with a high melting point is a network solid

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	_____	

4.

A teacher candidate builds a worked example for $\text{H}_2 + \text{Cl}_2 \rightarrow 2\text{HCl}$, a reaction she saw in a video from a chemical plant. Bond energies: H-H 436 kJ/mol, Cl-Cl 242 kJ/mol, H-Cl 431 kJ/mol. What is ΔH for the reaction?

- A. +184 kJ
- B. +247 kJ
- C. -184 kJ
- D. -1,540 kJ

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

5. [TRANSFER — reversed stem / different variable]

A reaction breaks bonds totaling 1,000 kJ and forms bonds totaling 850 kJ. How is it classified?

- A. Exothermic, because breaking 1,000 kJ of bonds releases energy
- B. Exothermic, $\Delta H = -150$ kJ
- C. Endothermic, $\Delta H = +150$ kJ
- D. Endothermic, $\Delta H = +1,850$ kJ

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

6.

During a demonstration that the principal is scheduled to observe, a teacher candidate heats steel wool in an open dish on a digital balance. Most of the class predicted the mass would drop, as it does when paper burns. The mass rises. What is the best explanation to give?

- A. The heat energy added mass to the steel wool.
- B. Iron combined with oxygen from the air to form iron oxide, so atoms were added to the dish.
- C. The mass must stay the same, so the reading is an error.
- D. Burning always destroys some matter, and the balance is faulty.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

7. [TRANSFER — reversed stem / different variable]

10.0 g of calcium carbonate is heated in an open crucible until only 5.6 g of calcium oxide remains. How much carbon dioxide left the crucible?

- A. 0 g, because mass is always conserved
- B. 15.6 g
- C. 4.4 g
- D. 5.6 g

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

8.

A science-and-society lesson uses a local-news-style case: a pond near an invented factory town had fish die-offs during a heat wave in August, and a new parking lot was built nearby that spring. Which chemistry idea best explains the die-offs?

- A. Warmer water dissolves more of every substance, including oxygen.
- B. Runoff from the parking lot raised the water pressure.
- C. Oxygen reacted with the water to form hydrogen peroxide.
- D. Warmer water holds less dissolved oxygen.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

9.

A teacher candidate prepares four solutions for a boiling-point station: 1.0 m glucose, 1.0 m NaCl, 1.0 m CaCl_2 , and 1.5 m sucrose. Sucrose has the largest molar mass of the group. Which solution has the highest boiling point?

- A. 1.0 m CaCl_2
- B. The sucrose, because it has the largest molar mass
- C. 1.5 m sucrose
- D. 1.0 m NaCl

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

10. [TRANSFER — reversed stem / different variable]

A 1.0 m aqueous solution raises the boiling point of water by $1.54\text{ }^\circ\text{C}$ ($K_b = 0.512\text{ }^\circ\text{C/m}$). Which solute was used?

- A. Glucose
- B. NaCl
- C. AlCl_3
- D. CaCl_2

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

11.

A design challenge asks teams to build a water filter that removes at least 90% of the cloudiness from a sample and costs under \$5 in materials. Teams of four meet in the library on Thursdays. Which part of the task is a constraint?

- A.** Explaining why sand traps particles
- B.** Costs under \$5 in materials
- C.** Removes at least 90% of the cloudiness
- D.** Teams of four meeting on Thursdays

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

12.

During a unit on gases, in a class that recently switched to block scheduling, a learner says, "Kinetic molecular theory is still just a theory; once it is proven, it will become a law like Boyle's law." What is the most accurate response?

- A.** Yes, with enough evidence the theory will be promoted to a law.
- B.** A theory is an educated guess that has not been tested yet.
- C.** Laws are never revised, while theories change often.
- D.** Boyle's law describes how pressure and volume relate; kinetic molecular theory explains why. A theory does not turn into a law.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

STOP

Answers and dissection follow. Finish every item and every strip first.

Key and dissection

Struck-through text is the camouflage: facts that do not decide the item.

1. Answer B Concept 1

While planning an atomic-structure lesson ~~for a tenth-grade class that meets after lunch~~, a teacher candidate uses boron: 20.0% is boron-10 (10.0 u) and 80.0% is boron-11 (11.0 u). ~~The periodic table poster in the room is faded.~~ What average atomic mass should the answer key show?

Target word: weighted average atomic mass **Decides:** Weight each mass by its abundance.

A	Confusion set	Reports only the most abundant isotope.
B	Correct	$0.200 \times 10.0 + 0.800 \times 11.0 = 2.0 + 8.8 = 10.8$ u.
C	Inverted operation	Swaps the two abundances.
D	Dropped element	Simple average; leaves out the abundances.

2. Answer A Concept 1 • transfer

An element has two isotopes, 63 u and 65 u, and an average atomic mass of 63.5 u. What percent of its atoms are the 63 u isotope?

Target word: solve for abundance **Decides:** $63x + 65(1 - x) = 63.5$.

A	Correct	$65 - 2x = 63.5 \rightarrow x = 0.75$.
B	Absolute language	Assumes isotopes are always equally common.
C	Wrong stem number	Reuses the average mass as a percent.
D	Inverted operation	That is the abundance of the 65 u isotope.

3. Answer A Concept 2

In a lab station ~~set up next to the room's only sink~~, a white crystalline solid melts at 801 °C. It does not conduct electricity as a solid, but it conducts well when dissolved in water. ~~The solid came from a kit ordered from a discount supplier.~~ What type of bonding does it have?

Target word: bond type **Decides:** High melting point; conducts only when ions are free to move.

A	Correct	Ions are locked in the solid and mobile in solution.
B	Adjacent category	Molecular solids melt low and their solutions usually do not conduct.
C	Confusion set	Metals conduct as solids too.
D	Absolute language	Network solids do not conduct when dissolved; high melting point alone does not decide.

4. Answer C Concept 3

A teacher candidate builds a worked example for $\text{H}_2 + \text{Cl}_2 \rightarrow 2\text{HCl}$, ~~a reaction she saw in a video from a chemical plant.~~ Bond energies: H-H 436 kJ/mol, Cl-Cl 242 kJ/mol, H-Cl 431 kJ/mol. What is ΔH for the reaction?

Target word: ΔH from bond energies **Decides:** Broken: $436 + 242 = 678$. Formed: $2 \times 431 = 862$.

A	Inverted operation	Formed – broken reverses the sign.
B	Dropped element	Counts only one H-Cl bond formed.
C	Correct	$678 - 862 = -184$ kJ; more energy is released forming bonds than absorbed breaking them.
D	Confusion set	Adds all bond energies as if every bond released energy.

5. Answer C Concept 3 · transfer

A reaction breaks bonds totaling 1,000 kJ and forms bonds totaling 850 kJ. How is it classified?

Target word: sign and label from bond totals **Decides:** $\Delta H = 1,000 - 850 = +150$ kJ.

A	Confusion set	Bond breaking always absorbs energy.
B	Inverted operation	Subtracts in the wrong order.
C	Correct	More energy absorbed breaking bonds than released forming them.
D	Adjacent category	Adds the totals instead of taking the difference.

6. Answer B Concept 4

During a demonstration ~~that the principal is scheduled to observe~~, a teacher candidate heats steel wool in an open dish on a digital balance. Most of the class predicted the mass would drop, ~~as it does when paper burns~~. The mass rises. What is the best explanation to give?

Target word: why an open-system mass increased **Decides:** Iron combines with oxygen from the air, adding mass.

A	Adjacent category	Heat energy has no measurable mass at this scale.
B	Correct	Mass is conserved overall; oxygen entered the open system.
C	Absolute language	Mass stays constant only in a closed system.
D	Confusion set	Matter is never destroyed in chemical reactions.

7. Answer C Concept 4 · transfer

10.0 g of calcium carbonate is heated in an open crucible until only 5.6 g of calcium oxide remains. How much carbon dioxide left the crucible?

Target word: mass of the escaped gas **Decides:** Mass conserved: reactant mass – solid product mass.

A	Absolute language	Mass is conserved in total, but the gas left the open crucible.
B	Inverted operation	Adds instead of subtracting.
C	Correct	$10.0 - 5.6 = 4.4$ g of CO ₂ .
D	Wrong stem number	The mass of the solid product.

8. Answer D Concept 5

A science-and-society lesson uses a local-news-style case: a pond ~~near an invented factory town~~ had fish die-offs during a heat wave in August, ~~and a new parking lot was built nearby that spring~~. Which chemistry idea best explains the die-offs?

Target word: why warm water holds less oxygen **Decides:** Gas solubility falls as water temperature rises.

A	Absolute language	Many solids dissolve better when warm, but gases do the opposite.
B	Camouflage	The parking lot is not the deciding factor here; temperature is.
C	Confusion set	Dissolved O ₂ does not react with water this way.
D	Correct	Gases escape more easily from warm water.

9. Answer A Concept 6

A teacher candidate prepares four solutions for a boiling-point station: 1.0 m glucose, 1.0 m NaCl, 1.0 m CaCl₂, and 1.5 m sucrose. ~~Sucrose has the largest molar mass of the group.~~ Which solution has the highest boiling point?

Target word: highest particle concentration **Decides:** $i \times m$: glucose 1.0, NaCl 2.0, CaCl₂ 3.0, sucrose 1.5.

A	Correct	Each unit gives 3 ions, for 3.0 m of particles.
B	Camouflage	Colligative properties depend on particle count, not molar mass.
C	Dropped element	Highest molality, but sucrose does not dissociate, so only 1.5 m of particles.
D	Confusion set	NaCl gives 2 particles per unit, fewer than CaCl ₂ .

10. Answer D Concept 6 · transfer

A 1.0 m aqueous solution raises the boiling point of water by 1.54 °C ($K_b = 0.512$ °C/m). Which solute was used?

Target word: identify solute from i **Decides:** $i = 1.54 / (0.512 \times 1.0) \approx 3$.

A	Dropped element	$i = 1$ would give 0.512 °C.
B	Confusion set	$i = 2$ would give about 1.02 °C.
C	Adjacent category	$i = 4$ would give about 2.05 °C.
D	Correct	Ca ²⁺ + 2Cl ⁻ gives $i = 3$.

11. Answer B Concept 7

A design challenge asks teams to build a water filter that removes at least 90% of the cloudiness from a sample and costs under \$5 in materials. ~~Teams of four meet in the library on Thursdays.~~ Which part of the task is a constraint?

Target word: identify the constraint **Decides:** Constraints are limits; the cost cap limits the design.

A	Adjacent category	That is a science explanation, not a design limit.
B	Correct	A budget is a limit on possible solutions.
C	Confusion set	That is a criterion: the performance the design must reach.
D	Camouflage	Classroom logistics, not a limit on the filter design.

12. Answer D Concept 8

During a unit on gases, ~~in a class that recently switched to block scheduling,~~ a learner says, “Kinetic molecular theory is still just a theory; once it is proven, it will become a law like Boyle’s law.” What is the most accurate response?

Target word: relationship between laws and theories **Decides:** Laws describe; theories explain. They are different kinds of knowledge.

A	Confusion set	This is the common misconception the item targets.
B	Adjacent category	That describes a hypothesis, not a theory.
C	Absolute language	Laws can be refined or limited by new evidence too.
D	Correct	Both are well supported; they do different jobs.

Triage grid

Write each item number in the box it belongs to.

	Right	Wrong
Sure	Owled.	Concept inversion — top priority. A wrong mental model; it survives rewording.
Unsure / guess	Recognition, not mastery. Re-drill with the transfer.	Not learned yet.

Trap tally

For every miss, find the letter you chose in the key and tally its trap. The biggest number is the pattern to drill.

Trap	What it does	In this drill	Your misses
Dropped element	Leaves out a required term or step.	4	
Wrong stem number	A figure copied from the prompt that is not the answer.	2	
Adjacent category	The right idea in the wrong bucket.	6	
Inverted operation	Direction flipped: add/subtract, cause/effect, higher/lower.	5	
Absolute language	Always, never, guarantee — where judgment applies.	6	
Confusion set	Every option is real; the test is exact role-matching.	10	
Camouflage	Answers a stem fact that was never the question.	3	

Anchor rows — the concept list

One row per concept: the rule to lock in, the trap that catches people, and the tell in the question that signals it.

#	Concept	Rule	Trap	Tell
1	Weighted average atomic mass	Average atomic mass = $\Sigma(\text{isotope mass} \times \text{fractional abundance})$, so it sits closest to the most abundant isotope.	Taking a simple average of the isotope masses.	Isotope masses with percent abundances.
2	Bond type from properties	Ionic solids have high melting points and conduct only when melted or dissolved; metals conduct as solids; molecular solids melt low and do not conduct.	Thinking any conducting substance is a metal.	Conductivity reported as solid vs. molten or dissolved.
3	Bond energy and ΔH	Breaking bonds absorbs energy and forming bonds releases it; $\Delta H \approx \Sigma(\text{bonds broken}) - \Sigma(\text{bonds formed})$.	Believing that breaking a bond releases energy.	A table of bond energies.
4	Conservation of mass, open vs. closed	Mass is conserved in every reaction; an open-system mass change means a gas entered or left.	Concluding that matter was created or destroyed when a balance reading changes.	A reaction on a balance in an open dish, or gas produced.
5	Gas solubility and temperature	Gases become less soluble in water as temperature rises, the opposite of most solids.	Assuming every solute dissolves better when warm.	Dissolved oxygen, carbonation, or warm water.
6	Colligative properties count particles	Boiling-point elevation and freezing-point depression depend on moles of dissolved particles ($i \times m$), not on the identity or molar mass of the solute.	Ignoring that ionic solutes split into several particles.	Several solutions at given molality compared for boiling or freezing point.
7	Engineering criteria vs. constraints	Criteria are the goals a design must meet; constraints are limits such as cost, time, materials, or safety.	Calling a performance goal a constraint, or treating design work as ordinary inquiry.	A design task with a budget and a performance target.
8	Scientific law vs. theory	A law describes a pattern (what happens); a theory explains why. Theories do not graduate into laws.	Teaching that a theory becomes a law once it is proven.	A learner says a theory is “just a guess” or will “become a law.”

Course scope (public description): Secondary Chemistry Curriculum synthesizes the core content of the Secondary Chemistry Program, preparing candidates to apply education theory and teaching approaches to the chemistry content areas. This course aligns with the areas covered by the Chemistry Praxis exam, ensuring candidates demonstrate both a deep understanding of the subject matter and the ability to effectively deliver content in a secondary classroom setting. Topics include atomic structure, chemical bonding, energy relationships, chemical reactions, solutions, and the role of science and engineering in society. There are no prerequisites for this course.